Sentiments

A Treatise on the Origins of Fluctuations

Angeletos & La'O *Econometrica* 2013

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New and Old Ideas in Macroeconomics

PSE April 8th 2026

Today's discussion

1 The theoretical context

2 An "island model" with a Keynesian flavour

3 The anatomy of extrinsic fluctuations

4 Sentiment augmented RBCs

The ingredients of modern macroeconomics

1. Dynamic optimisation
2. Rational expectations
3. A range of frictions $\pm$ Walrasian market clearing

These concepts came about mostly as a reponse to reduced-form large-scale macroeconometric "Keynesian" models $\rightsquigarrow$ Phillips curve vs. Natural Rate Hypothesis

$\hookrightarrow$ Reduced form parameters are rationalized as functions of the structural parameters that govern (1)-(3). A form of scientific reductionism? *De facto* a largely operational framework

Example: Lucas (1972)

This paper provides a simple example of an economy in which equilibrium prices and quantities exhibit what may be the central feature of the modern business cycle: a systematic relation between the rate of change in nominal prices and the level of real output. The relationship, essentially a variant of the well-known Phillips curve, is derived within a framework from which all forms of "money illusion" are rigorously excluded: all prices are market clearing, all agents behave optimally in light of their objectives and expectations, and expectations are formed optimally (in a sense to be made precise below).

Rational Expectations Equilibrium

The novelty that Rational Expectations introduce is that the equilibrium object is a function defined on the state space of the model.
$\hookrightarrow$Find the decision rule/price which induces dynamic paths such that the decision rule is indeed optimal given this path and such that the actual and expected path coincide on average.

DEFINITION. An equilibrium price is a continuous, nonnegative function $p(\cdot)$ of (m, x, θ), with $mx/\theta p(m, x, \theta)$ bounded and bounded away from zero, which satisfies:

$$h\left[\frac{mx}{\theta p(m, x, \theta)}\right]\frac{1}{p(m, x, \theta)} = \int V'\left[\frac{mxx'}{\theta p(m\xi, x', \theta')}\right]\frac{x'}{p(m\xi, x', \theta')}\, dG(\xi, x', \theta' \mid p(m, x, \theta)). \quad (4.2)$$

What about business cycle fluctuations?

If the path induced by the solution is stable, the only thing that can account for fluctuations are parametric changes of that solution $\rightsquigarrow$ exogenous shocks

- Technology
- Preferences
- Policy rules

Apart from well identified cases (monetary policy shocks, oil shocks), these are largely counter-factual: quarterly changes in the technological frontier, or the willingness to consume/work.

If you disagree with this, there are (roughly) three ways forward:

- Yield a unique but unstable solution (*e.g.* Grandmont (1985), Benhabib and Nishimura (1979), Beaudry, Portier and Galizia (2020)). Not for today
- Exhibit a whole family of solutions: indeterminacy
- Depart from rationality

Sunspots

If there are many solutions to the functional equation that defines the REE, fluctuations may be accounted for by random shifts across the solution space.

Which ones?

Need for a common randomization device:

If $E_1, E_2, ...E_n$ are the different solutions, then postulate a r.v. $s \in S$ and a rule $g : s \mapsto E(s)$.

The r.v. s is called a **sunspot**. A sunspot equilibrium is thus constructed as a randomization over fundamental equilibria (Cass & Shell 1983, Azariadis & Guesnerie 1986).

$\hookrightarrow$ How to make quantitative predictions? On what ground can we choose the sunspot variable and the rule g?

Where this paper sits

A bit in between:

- Unique stable REE with "sentiment" shocks
  - Waves of exhuberance and pessimims $\perp\!\!\!\perp$ fundamentals $\approx$ "animal spirits"
  - No departure from rationality

How come?

- Lack of a central market mechanism $\rightarrow$ imperfect communication between "islands"

  $\hookrightarrow$ no agents reach the same expectation. Related to global games.

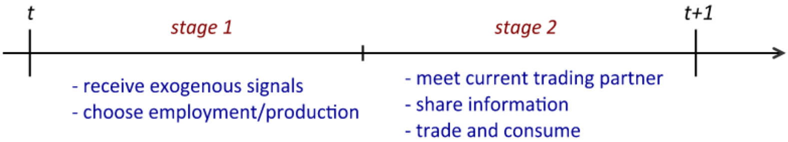
- Aggregate fluctuations in these expectations may be driven by an extrinsic shock

Model ingredients

- Continuum of islands, with a representative household and firm
- The "Home" good can be consumed on the island or traded for a "foreign" good
- CRS production using fixed supply of land and elastic labour, TFP varies across islands
- Trading takes place through random pairwise matching (iid uniform)
- Witihin a period, the employment and production decision is made before trading and consumption take place
    - Decisions are made in anticipation of future trading opportunities, and with incomplete information about these opportunities.

      $\hookrightarrow$ *"Salto mortale"* $C - M'$ (but no money...)
    - Receive exogenous signals when they make their decisions, and exchange information when they trade

Timing


t
stage 1
stage 2
t+1
- receive exogenous signals
- choose employment/production
- meet current trading partner
- share information
- trade and consume

Model Ingredients (technicalities)

- Production:$y_i = A_i(n_{it}^{\theta})(K^{1-\theta})$
- Preferences: $\mathcal{U}_i = \sum_{t=0}^{\infty} \beta^t U(c_{it}, c_{it}^*) - V(n_{it})$ with $U(c, c^*) = \left(\frac{c}{1-\eta}\right)^{1-\eta} \left(\frac{c^*}{\eta}\right)^{\eta}$ and $V(n) = \frac{n^{\varepsilon}}{\varepsilon}$
- Budget constraint: $p_i c_{it} + p_i^* c_{it}^* \leq w_{it} n_{it} + r_{it} K + \pi_{it}$
- Information structure: Receive signal x_{it} a random variable drawn from $\mathcal{X}_t \subset \mathbb{R}^n$ compact. Information state:
  - Arrival of new signals $\omega_{it} = (z_{i,t-1}, x_{it})$
  - Endogenous information exchange $z_{it} = (\omega_{it}, \omega_{m(i),t})$

Joint distribution of signals x_{it} depends on an **exogenous random variable ξ_t**

Equilibrium

The underlying probability space of our model is quite rich, as it involves the realizations of all matches and signals in the population. For our purposes, however, it suffices to focus on the joint distribution of the history ξ^t of the sentiment shock and of the pair of information sets $(\omega_{it}, \omega_{jt})$ of an arbitrary match (i, j). We assume that this distribution is represented by a continuous probability density function, which we henceforth denote by $\mathcal{P}_t(\omega_{it}, \omega_{jt}, \xi^t)$. Next, note that any allocation and price system can be represented with a collection of functions $\{n_t, k_t, y_t, w_t, r_t, p_t, p_t^*, c_t, c_t^*\}_{t=0}^{\infty}$ such that, for all islands, dates, and possible states, $n_{it} = n_t(\omega_{it})$, $k_{it} = k_t(\omega_{it})$, $y_{it} = y_t(\omega_{it})$, $w_{it} = w_t(\omega_{it})$, $r_{it} = r_t(\omega_{it})$, $p_{it} = p_t(z_{it})$, $p_{it}^* = p_t^*(z_{it})$, $c_{it} = c_t(z_{it})$, and $c_{it}^* = c_t^*(z_{it})$, with $z_{it} = (\omega_{it}, \omega_{jt})$ and $j = m_t(i)$.[7] We require that these functions be continuous; this guarantees that all relevant expectations are well defined and permits us

An equilibrium is a collection of continuous allocation and price functions such that (i) given current prices and expectations of future prices, the allocations are optimal for households and firms; (ii) prices clear all markets; and (iii) expectations are rational.

Equilibrium characterisation

The equilibrium production decision and terms of trade solve the following fixed-point problem

$$\boldsymbol{y}_t(\omega) = \left(\theta^{\vartheta}\boldsymbol{A_t}(\omega)K^{1-\theta}\right)^{\frac{1}{1-\vartheta}}\left(\int_{\Omega_t}\boldsymbol{p_t}(\omega,\omega')\mathcal{P}_t(\omega'|\omega)d\omega'\right)^{\frac{\vartheta}{1-\vartheta}}$$

$$\boldsymbol{p}_t(\omega,\omega') = \boldsymbol{y_t}(\omega)^{-\eta}\boldsymbol{y_t}(\omega')^{\eta}$$

$\hookrightarrow$ CMT$\rightarrow$ uniqueness. Can also be seen as the PBE of a game with best response:

$$\log(y_{it}) = (1-\alpha)f_i + \alpha\mathbf{E}_{it}\left[\log(y_{jt})\right]$$

Where f_i summarizes the fundamentals and $\alpha = \frac{\eta}{\eta+(1-\vartheta)/\vartheta}$ pins down the degree of strategic complementarity

But the interdependence of incentives comes from the general equilibrium interactions, and are not hard-wired into the model ($\approx$).
$\hookrightarrow$ Firms entertains beliefs about other firms actions because they care about prices

Imperfect communication is the key

Can fluctuations in ξ_t drive fluctuations in Y_t?

- Perfect communication: two islands within a match share the same belief about each other's output level

  $\hookrightarrow$ The equilibrium is pinned down by fundamentals

- Imperfect communication: fluctuations can only be driven by changes in ξ_t

Example: $x_{it} = (x^1_{it}, x^2_{it})$ with $x^1_{it} = \log(A_j) + u^1_{it}$ and $x^2_{it} = x^1_{jt} + \xi_t + u^2_{it}$, $\xi_t \perp\!\!\!\perp$ fundamentals, but represents an aggregate noise component in "communication". With a few normalizations and assumptions we have:

$\log(Y_t)$ and $\log(\mathcal{B}_t)$ are increasing linear functions of ξ_t. At the micro level:

$$\log y_{it} = \phi_0 + \phi_a \log A_i + \phi_1 x^1_{it} + \phi_2 x^2_{it}$$
$$\mathbb{E}_{it} \log p_{it} = \psi_0 - \psi_a \log A_i + \psi_1 x^1_{it} + \psi_2 x^2_{it}$$

Higher-order beliefs?

Imperfect communication means that firms lack some common knowledge : Even if firm i knows firm j's productivity, it doesn't know if j knows that it knows, and if j knows that it knows that it knows and so on... $\rightarrow$ **Higher-order beliefs: the forecast of the forecast of others.**

The shock ξ_t can be understood as a shock to higher-order beliefs of exogenous fundamentals.

Footnote: Early RE justification is precisely this "eductive" activity (Guesnerie 1992)

Isn't this complicated? And unrealistic?

No, because these shocks to higher order beliefs are equivalent to first-order shocks of endogenous economic outcomes.

$\hookrightarrow$ In a RE equilibrium model, what matters is the joint distribution of equilibrium expectations and equilibrium outcomes.

Higher-order beliefs, cont.

$$x_{it}^1 = \log(A_j) + \varepsilon_{it}^1 \text{ and } x_{it}^h = x_{jt}^{h-1} + u_{it}^h + \xi_t^h, \ \forall h \in \{2, \ldots, H\}$$

$\hookrightarrow$ islands receive signals of the signals of the signals. . . of the signals of others. Shocks ξ_t^h imply changes in beliefs of order $\geq h$

In equilibrium we have:

$$\log(Y_t) = \Phi \bar{a}_t + \psi \bar{\xi}_t \text{ and } \log(\mathcal{B}_t) = \Phi \bar{a}_t + \bar{\xi}_t$$

where $\bar{\xi}_t = \sum \lambda_h \xi_t^h \rightarrow$ a single "composite shock" matters.

$\hookrightarrow$Roughly put, since agents are only interacting through prices, all they really care about are prices, so that shocks that x-th order beliefs are equivalent to shocks about expectations of prices directly and therefore not particularly relevant

Why "sentiments" then?

Higher-order beliefs shocks have the same empirical content as shocks to equilibrium expectations

$\hat{\mathbb{E}}_{it} \log(y_{jt}) = \log y^*_{jt} + \zeta_t$ where y^*_{jt} is the FIRE benchmark and ζ_t represents an irrational shift to "market psychology"

Key result: the equilibrium (given this belief specification) is the same as when there are shocks to higher-order beliefs:

$$\log(Y_t) = \Phi \bar{a}_t + \psi \zeta_t \text{ and } \log(\mathcal{B}_t) = \Phi \bar{a}_t + \zeta_t$$

Note that this is not a RE model, BUT it is observationally equivalent to a model with sentiment shocks

$\hookrightarrow$ Sentiment shocks rationalize exactly the seemingly random and irrational shifts in expectations of economic activity

Which informaion friction nests the result?

The model definitely has a flavour of coordination failures because of the trading frictions that impinge equilibrium communication

$\hookrightarrow$ Coordination failures through strategic complementarities and multiple equilibria have been used to model this Keynesian insight (Diamond 1982, Cooper & John 1997)

What is driving the equilibrium volatility here?

$$y_i = (1-\alpha)\mathbb{E}_i[A] + \alpha\mathbb{E}_i(Y) \Longrightarrow Y = (1-\alpha)\sum_h \alpha^{h-1}\bar{\mathbb{E}}^h[A].$$

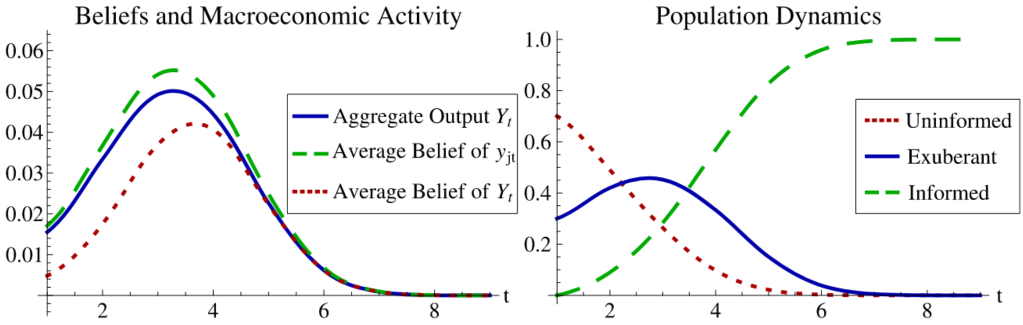
Since $\mathbb{V}(\bar{\mathbb{E}}^h[A]) \leq \mathbb{V}(\bar{\mathbb{E}}^{h+1}[A])$ we have $\mathbb{V}(Y) \leq \mathbb{V}(A)$

$\hookrightarrow$Incomplete information about fundamentals cannot give rise to excess volatility.

Contrast with $\log(y_{it}) = (1-\alpha)f_i + \alpha\mathbf{E}_{it}\left[\log(y_{jt})\right] \rightarrow$ forecast of a random tranding partner//The aggregate action. Therefore, can sustain aggregate volatility from sufficiently large idiosyncratic risk and sufficiently correlated beliefs about those

Communication and boom busts cycles

Similar setup as before but with two groups and two subgroups: North and South, half of which get a partial signal about the other group's productivity. Only one-off signal, but as islands meet, they share information.



Wave of optimism then correction drives fluctuations in GDP.

Adding dynamics: model setup

RBC+island model: goods are traded across islands are intermediate inputs for the production of a local final good for investment or consumption:

Final good market clearing:$y_{it} = c_{it} + i_{it}$

Capital accumulation with convex capital utilization costs: $k_{it} = (1 - \Delta(e_{i,t}))k_{it} + i_t$

Production of the local input: $q_{it} = A_i(e_{it}k_{it})^{1-\theta}(n_{it})^{\theta}$

Production of the final good:$y_{it} = \frac{1}{\zeta}h_{it}^{1-\eta}(h_{it}^{*})^{1-\eta}$

Market clearing for intermediate inputs $q_{it} = h_{it} + h_{jt}^{*}$

Preferences are the same as before

Equilibrium characterisation

$$
\begin{aligned}
V'(n_{it}) &= \theta\, \xi\, \mathbb{E}_{it}\left[U'(c_{it}) \frac{y_{it}}{n_{it}} \mid \omega_{it} \right] \\
\Delta'(e_{it}) e_{it} &= (1-\theta)\, \zeta\, \mathbb{E}_{it}\left[U'(c_{it}) \frac{y_{it}}{k_{it}} \mid \omega_{it} \right] \\
U'(c_{it}) &= \beta\, \mathbb{E}_{it}\left[U'(c_{i,t+1}) \left(1 + (1-\theta)\, \xi\, \frac{\mu}{1+\mu} \frac{y_{i,t+1}}{k_{i,t+1}} \right) \mid z_{it} \right] \\
c_{it} + k_{i,t+1} &= y_{it} + \left(1 - \Delta(e_{it})\right) k_{it} \\
y_{it} &= q_{it}^{1-\eta} q_{jt}^{\eta} \\
q_{it} &= A_i \,(e_{it} k_{it})^{1-\theta} (n_{it})^{\theta}
\end{aligned}
$$

Again PBE analogy: it is *as if* the islands are playing a dynamic game in which an island's optimal employment, consumption, and investment choices during one period depend on the expected output of its likely trading partner, not only in the current period, but also in all future periods.

Solving the model

Problem: there is an infinite-regress in forecasting the forecast of other agents (Townsend 1983).

$\hookrightarrow$ Need a infinite dimensional state space: an expectation j of the expectation i of the expectation j of what firm i is doing and so on and so on...

Similar idea as the Master Equation in HA models: a distribution of beliefs is necessary for inference about an aggregate variable (prices/output), which depends on the distribution of beliefs.

Solution: heterogenous priors$\rightarrow$ each island receives a single signal about its trading partner $x_{it} = \log(A_{j,t}) + \varepsilon_{it}$

- Each island believes that its own error is unbiased $\mathcal{N}(0, \sigma_\varepsilon^2)$
- And that the errors of all other islands are biased $\mathcal{N}(\xi_t, \sigma_\varepsilon^2)$ where $\xi_t \sim$ Markov

If ξ_t is CK, then the (log-linearized) solution to the model can be represented as:

$$\tilde{K}_{t+1} = \Gamma(\xi_t, \tilde{K}_t)$$

(Under the hood: K_t is the best forecast of $k_{j,t}$ because A_j i.i.d. over time and across islands, hence the problem dumbs down to a fixed point or consistency condition between individual and aggregate outcomes)

Moments

TABLE I

BUSINESS-CYCLE STATISTICS OF OUR MODEL ALONG WITH THOSE OF THE U.S. ECONOMY[a]

	The Model		U.S. Data	
	std. dev	corr(X, Y)	std. dev	corr(X, Y)
Y	1.67	1.00	1.67	1.00
N	1.41	1.00	1.43	0.87
C	1.21	0.98	1.27	0.80
I	4.14	0.96	5.48	0.82
Y/N	0.26	0.99	0.82	0.51
LW	4.95	−1.00	4.98	−0.82

La Fin